一、单项选择题(从下列各题四个备选答案中选出一个答案,并将其字母代号写在该题的【　　】内。答案错选或未选者，该题不得分。每小题 3 分，共 **15** 分。)

1. B

2. C

3. D

4. B

5. A

二、代数计算（本大题满分15分。）

```
syms a11 a12 a21 a22 b11 b12 b21 b22
A = [a11 a12; a21 a22];
B = [b11 b12; b21 b22];
B_inv = inv(B);
AB = A - B_inv;
AB_inv = inv(AB);
A_inv = inv(A);
left_side = AB_inv - A_inv;
ABA = A*B*A;
right_side = ABA - A;
right_side = simplify(right_side);
%isEqual = isAlways(left_side==right_side)直接判断无法实现
simplify(left_side*right_side)
```

(2 分) **(5 分)** **(3 分)** **(5 分)**

ans =

[1, 0]
[0, 1]

三、微积分计算（本大题满分 15 分。）

解：已知$f(x,y)=\frac{ax^2+e^y sin3y}{by^2}$.

```
syms a b x y
f=(a*x^2+exp(y)*sin(3*y))/(b*y^2);%注意不能写成了 f=(a*x^2+sin(3y))/b*y^2;
fx=diff(f,x)
```

fx =

(2*a*x)/(b*y^2)　　**(5 分)**

```
fy=diff(f,y)
```

fy =

(3*cos(3*y)*exp(y) + sin(3*y)*exp(y))/(b*y^2) - (2*(a*x^2 + sin(3*y)*exp(y)))/(b*y^3)　　**(5 分)**

```
intx=int(f,x)
```

intx =

(x*(a*x^2 + 3*sin(3*y)*exp(y)))/(3*b*y^2)+C **(5 分)**

或

```
from sympy import *
a,b,x,y=symbols('a b x y')
f=(a*x**2+exp(y)*sin(3*y))/(b*y**2)
dfdx= diff(f,x)
dfdy= diff(f,y)
intf=integrate(f,x)
print('df/dx=',dfdx)
print('df/dy=',dfdy)
print('积分为{}+Const'.format(intf))
```

df/dx= 2*a*x/(b*y**2) **(5 分)**

df/dy= (exp(y)*sin(3*y) + 3*exp(y)*cos(3*y))/(b*y**2) - 2*(a*x**2 + exp(y)*sin(3*y))/(b*y**3) **(5 分)**

积分为 a*x**3/(3*b*y**2) + x*exp(y)*sin(3*y)/(b*y**2)+Const **(5 分)**

四、综合作图（本大题满分 15 分。）

```
t=0:0.05:2*pi;
a=1;(2 分)
subplot(1,2,1)
n=5;
x=a*sin(n*t).*cos(t);
 y=a*sin(n*t).*sin(t);
 plot(x,y)
 subplot(1,2,2)(5 分)
n=6;(3 分)
x=a*sin(n*t).*cos(t);
 y=a*sin(n*t).*sin(t);
 plot(x,y)(5 分)
```

或

```
import matplotlib.pyplot as plt
import numpy as np
ax1=plt.subplot(1,2,1)
a=1
n=7(2 分)
t=np.linspace(0, np.pi*2,101)
x=a*np.sin(n*t)*np.cos(t)
y=a*np.sin(n*t)*np.sin(t)
ax1.plot(x,y)
ax2=plt.subplot(1,2,2)(5 分)
a=1
n=8(3 分)
t=np.linspace(0, np.pi*2,101)
x=a*np.sin(n*t)*np.cos(t)
y=a*np.sin(n*t)*np.sin(t)
ax2.plot(x,y)
```

```
plt.show()
```
(5 分)

五、数据分析（本大题满分 15 分。）

```
% 成绩数据
scores = [23    88    14    64    47    94    72    18
          77    41    29    19    92    66    31    43
          83    16    52    70    13    59    86    27
          55    38    67    32    95    28    79    44
          91    60    49    73    82    17    33    51
          36    11    78    87    21    63    48    98];
```
(1 分)

```
% 初始化四个等级的数组
excellent = [];
good = [];
passing = [];
failing = [];
% 遍历成绩数据，将成绩划分到不同等级的数组中
```
(4 分)

```
for i = 1:size(scores, 1)
    for j = 1:size(scores, 2)
        score = scores(i, j);
        if 90 <= score && score <= 100
            excellent = [excellent, score];
        elseif 75 <= score && score < 90
            good = [good, score];
        elseif 60 <= score && score < 75
            passing = [passing, score];
        else
            failing = [failing, score];
        end
    end
end

% 对数组进行从大到小排序
excellent = sort(excellent, 'descend');
failing = sort(failing, 'descend');

% 输出每个等级从大到小排序后的数组
```
(5 分)

```
disp('优秀等级成绩（从大到小排序）：');
disp(excellent);
disp('不及格等级成绩（从大到小排序）：');
disp(failing);

% 初始化结果一维数组，用于存储每组的平均值
result = [];

% 计算每个等级数组的平均值并存储到 result 数组中
if ~isempty(good)
```

```
    result = [result, mean(good)];
end
if ~isempty(passing)
    result = [result, mean(passing)];
end

% 输出结果一维数组
disp('各等级平均值结果：');
disp(result);
```
(5 分)

或(Python)

```
scores = [[23, 88, 14, 64, 47, 94, 72, 18],
          [77, 41, 29, 19, 92, 66, 31, 43],
          [83, 16, 52, 70, 13, 59, 86, 27],
          [55, 38, 67, 32, 95, 28, 79, 44],
          [91, 60, 49, 73, 82, 17, 33, 51],
          [36, 11, 78, 87, 21, 63, 48, 98]]

# 初始化结果一维数组，用于存储每组的平均值，4 个元素（对应 4 个等级）
result = []

# 初始化四个等级的数组
excellent = []
good = []
passing = []
failing = []

# 遍历成绩数据，将成绩划分到不同等级的数组中
for row in scores:
    for score in row:
        if 90 <= score <= 100:
            excellent.append(score)
        elif 75 <= score < 90:
            good.append(score)
        elif 60 <= score < 75:
            passing.append(score)
        else:
            failing.append(score)
```
(5 分)

```
# 对每个等级的数组进行从大到小排序
excellent.sort(reverse=True)
failing.sort(reverse=True)

# 输出每个等级从大到小排序后的数组
print("优秀等级成绩（从大到小排序）：", excellent)
print("不及格等级成绩（从大到小排序）：", failing)
```
(5 分)

```
# 计算每个等级数组的平均值并存储到 result 数组中
if good:
    result.append(sum(good) / len(good))
if passing:
    result.append(sum(passing) / len(passing))

# 输出结果一维数组
print("中间等级平均值结果：", result)
```
(5 分)

六、优化计算（本大题满分 15 分。）

```
Max=2*x1+3*x2-x3+4*x4+5*x5-2*x6;
3*x1+2*x2-4*x3+x4+3*x5+2*x6<20;
X1-3*x2+2*x3-2*x4+4*x5-3*x6>10;
2*x1+4*x2+3*x3+5*x4-2*x5+x6=15;
X1+2*x2-x3+3*x4+2*x5-4*x6<10;
4*x1-2*x2+3*x3-2*x4-x5+3*x6<5;
@gin(x1);@gin(x3);@free(x4);
```
(2分)
(5分)
(3分)

```
 Global optimal solution found.
  Objective value:                              46.25000
  Extended solver steps:                              3
  Total solver iterations:                           14
                       Variable           Value        Reduced Cost
                             X1        0.000000            7.284091
                             X2        15.75000            0.000000
                             X3        8.000000           -0.5681818
                             X4       -10.50000            0.000000
                             X5        10.00000            0.000000
                             X6       0.5000000            0.000000
```

算出max=46.25 x1=0 x2=15.75 x3=8 x4=-10.5 x5=10 x6=0.5 **(5分)**

七、编程实现（本大题满分 10 分。）

```
function main7
[randle_nums, num_count] = find_randle_numbers(3);
disp(randle_nums);
disp(num_count);
end
function [randle_numbers, count] = find_randle_numbers(n)
    if n < 3 || n > 6
        error('输入的 n 需要满足 3 <= n <= 6');
    end
    start_num = 10^(n - 1);
    end_num = 10^n - 1;
```
(3 分)
(1 分)

```
    randle_numbers = [];
    count = 0;
    for num = start_num : end_num
        num_str = num2str(num);
        sum_power = 0;
        for i = 1 : n
            digit = str2double(num_str(i));
            sum_power = sum_power + digit ^ n;
        end
        if sum_power == num
            randle_numbers = [randle_numbers, num];
            count = count + 1;
        end
    end
end
```

(3 分)

n=3 时

153　370　371　407

4

n=4 时

1634　8208　9474

3

n=5 时

54748　92727　93084

3

n=6 时

548834

1

(3 分)